



Figure 2: 3 trends mapped across the Data-to-Policy Pipeline

Recommendations For Future Work

sensors carry a documented, physics-based measurement bias that current regulators do not mandate manufacturers to correct before market access.

Finalize Fitzpatrick-Scale Hardware Validation

finalize binding pre-market accuracy requirements for PPG devices specifying minimum thresholds across all Fitzpatrick categories, using arterial blood gas as ground truth for comparison.

Develop a Public PPG Fairness Benchmark

A public, demographically annotated PPG dataset should be developed under NIH or equivalent oversight to enable reproducible, comparable evaluation of bias mitigation techniques across research groups.

Subgroup Performance Reporting

Regulatory submissions for AI-enabled RPM tools should require disaggregated performance reporting using standard fairness metrics, not aggregate accuracy alone.